Christian Rafael de Souza Silva



Software Engineer — ML Systems, AI Infrastructure & Systems Software

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Software development since 2012, with professional experience in enterprise systems, retail and fiscal applications, APIs and desktop software. Independent projects focus on ML infrastructure and systems programming.

EXPERIENCE

Nov 2023 — Present

Information Systems Programmer — ITB Equipamentos Elétricos

Birigui, SP, Brazil

- Internal management system covering finance, accounting, production planning and engineering.
- C++, MySQL and Delphi; Horse APIs and Swagger documentation.
- .NET Core, EF Core, C#, JavaScript, React and PostgreSQL.
- DDD, CQRS, layered architecture and automated testing.
- Maintenance and extension of production systems, application APIs and automated testing.

Feb 2016 — Nov 2023

Full Stack Developer — Gigatron Franchising / Dotbank

Birigui, SP, Brazil

- PHP, Laravel, Angular, Vue and PostgreSQL; Android apps in Java/Kotlin and hybrid React/Ionic apps.
- APIs and products for ZIP code lookup, SMS, food delivery, retail automation, fiscal documents and ticketing.
- Linux/Windows Server administration; DigitalOcean, AWS and Cloudflare.
- Delphi application for production sheet stacking and printing.

Jul 2014 — Feb 2016

Full Stack Developer — Netstart Informática

Birigui, SP, Brazil

- Retail automation with Delphi/PostgreSQL and SAT fiscal queue processing with Redis.
- Desktop updater; PHP/CodeIgniter API and panel with jQuery and PostgreSQL.
- Internal tools, Bash scripts and cloud storage/querying of fiscal documents.

Jan 2014 — Jul 2014

Full Stack Developer — Gigatron Franchising

Birigui, SP, Brazil

- PHP/MySQL applications, server administration and SMS service using Python with a modem.

Aug 2012 — Jun 2013

Programmer Apprentice — Assessor Público (MV & P Tecnologia)

Araçatuba, SP, Brazil

- GeneXus applications, R&D, AWS server administration and OpenVPN.

Jan 2012 — Jul 2012

Programmer Apprentice — Gigatron Tecnologia

Birigui, SP, Brazil

- Internal applications in Delphi.

SELECTED PROJECTS

2025 —

ChrisOS

Experimental operating system and low-level playground for kernels, compilers, toolchains, virtual machines, emulators, drivers and graphics subsystems. GitHub · Documentation

2026

Chris-GPT-2

124M-parameter GPT-2 trained from scratch on ~10B FineWeb-Edu tokens using 4× A100 GPUs, with SafeTensors/GGUF exports. GitHub · Hugging Face

2026 —

Chris-Llama

Experimental LLM inference runtime in C covering GGUF, tokenization, attention, KV cache, quantization and CPU/GPU execution. GitHub

2026 —

Chris-Torch

Deep learning framework built from scratch with tensors, autograd, neural-network components, optimizers and accelerated backends. GitHub

2026 —

Velis

Collaborative workspace with device-local changes and multi-device synchronization. Product

2026 —

Chris Cleaner

C#/.NET desktop application for disk analysis, background file cataloging and user-controlled cleanup. Product

2026 —

Engineering Manual

Open, bilingual computer science and software engineering reference with full-text search, structured study tracks, editorial checks and automated static publishing. Website · GitHub

2026 —

Domine a Guitarra

Free interactive guitar-learning platform built with Astro and TypeScript, featuring chord diagrams, tablature playback, ear training and Web Audio synthesis. Website · Source

2026 —

SharpEx

Market and portfolio-intelligence workspace with multiple portfolios, risk/return metrics, crypto, market data and structural research integration.

2026 —

Marginal Thinking

Independent macroeconomics, markets and geopolitics research platform with recurring reports, structural datasets and primary-source analysis. marginalthinking.org

EDUCATION

- **Postgraduate Degree in Electronic Engineering & Robotics** — UNINTER, 2026
- **Associate Degree in Data Science** — UNINTER, 2023 — 2025
- **Bachelor's Degree in Electrical Engineering** — UNINTER, 2025 — 2030, in progress
- **Technical Degree in Mechatronics** — Colégio Salesiano Dom Lasagna, 2012 — 2014

TECHNICAL PROFILE

Languages: C, C++, Python, Rust, C#, Java, Kotlin, JavaScript/TypeScript, PHP, Delphi.

ML & systems: LLM training/inference, PyTorch, CUDA, distributed training, GGUF, quantization, Linux, operating systems, compilers, virtual machines and emulators.

Applications & infrastructure: .NET, React, Laravel, REST APIs, PostgreSQL, MySQL, Redis, DDD, CQRS, GitHub Actions, Cloudflare, AWS and DigitalOcean.